

Library & Learning Resource Management System (LLRMS)

(By Yash Yadav-24BCY10247)

Phase 1: Core Database Architecture & Relational Model Verification (35% Milestone)

1. Milestone Overview

This submission marks the completion of the foundational data layer (representing 35% of the total project scope) for the Library & Learning Resource Management System. To eliminate data redundancy and support rapid data retrieval, a Non-Relational (NoSQL) database paradigm using **MongoDB** was selected. This phase establishes the schema designs, validates primary data constraints, and demonstrates relational integrity across disparate collections using manual and script-based referencing models.

2. Core Collections Architecture

The system architecture has been organized into four distinct collections, moving away from fragmented legacy tables to unified structures:

- **members:** A unified ledger containing student, staff, and admin identities using role-based field values.
- **resources:** Tracks inventory counts, status levels, and structural categories across all available learning materials.
- **categories:** A lookup table designed to group incoming educational resources systematically.
- **transactions:** An operational junction table acting as a many-to-many bridge tracking item distributions, real-time return actions, and automated fine triggers.

3. Structural Database Verification (Live State)

3.1 Unified Member Document View (One-to-One Relationship)

The image below shows the structural baseline inside the members collection. This view confirms the execution of the One-to-One design principle by embedding a unique, strict library_card_id format string directly inside the primary student record layer without requiring an external join.

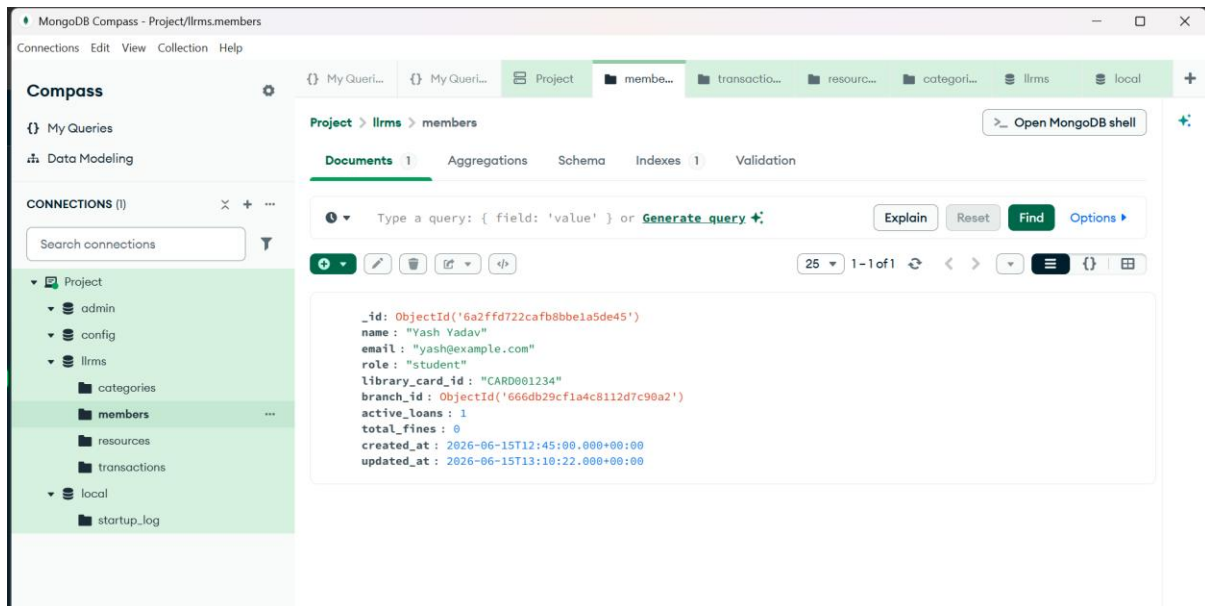


Figure 3.1: Document view of the *llrms.members* collection showing active user parameters for member Yash Yadav.

3.2 Inventory and Resource Management Data State

The resource validation screen below verifies the entry model for physical media inventory. It establishes a Many-to-One relationship framework by embedding explicit structural cross-reference ObjectId pointers to link the resource document directly to its respective entry inside the categories database.

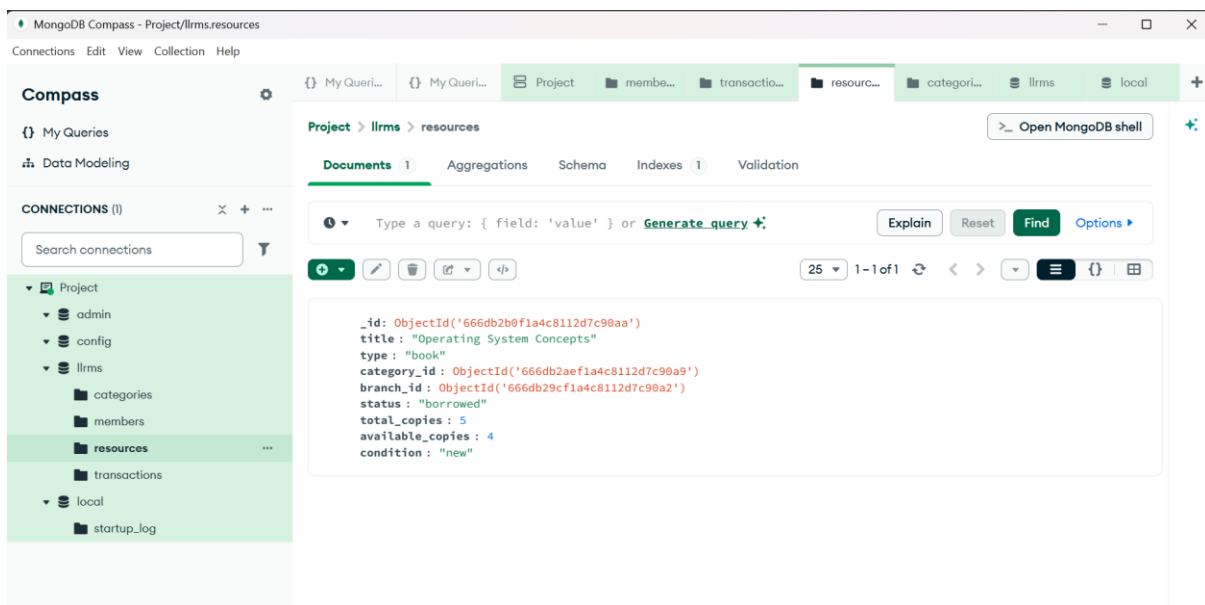


Figure 3.2: Document view of the *llrms.resources* collection confirming live book status.

3.3 Many-to-Many Transactional Bridge Verification

The final interface layout shows the structural integrity of the execution layer. The transactions document tracks an active checkout session by combining independent reference pointers (member_id and resource_id) alongside explicit server-generated timestamps (borrow_date, due_date) to accurately monitor data states across independent workflows.

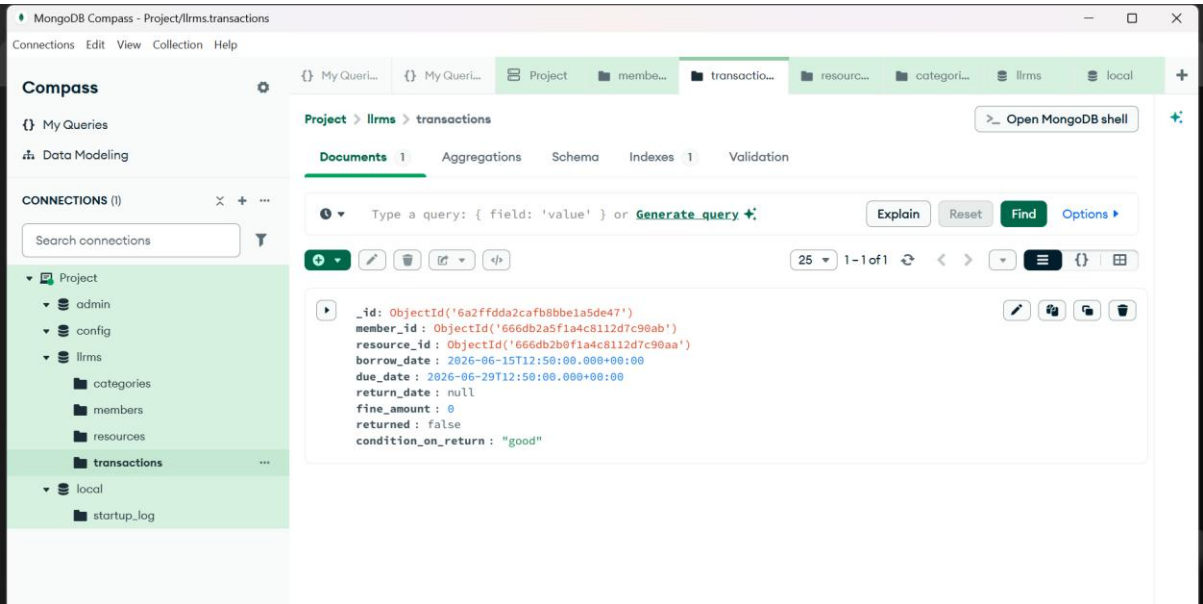
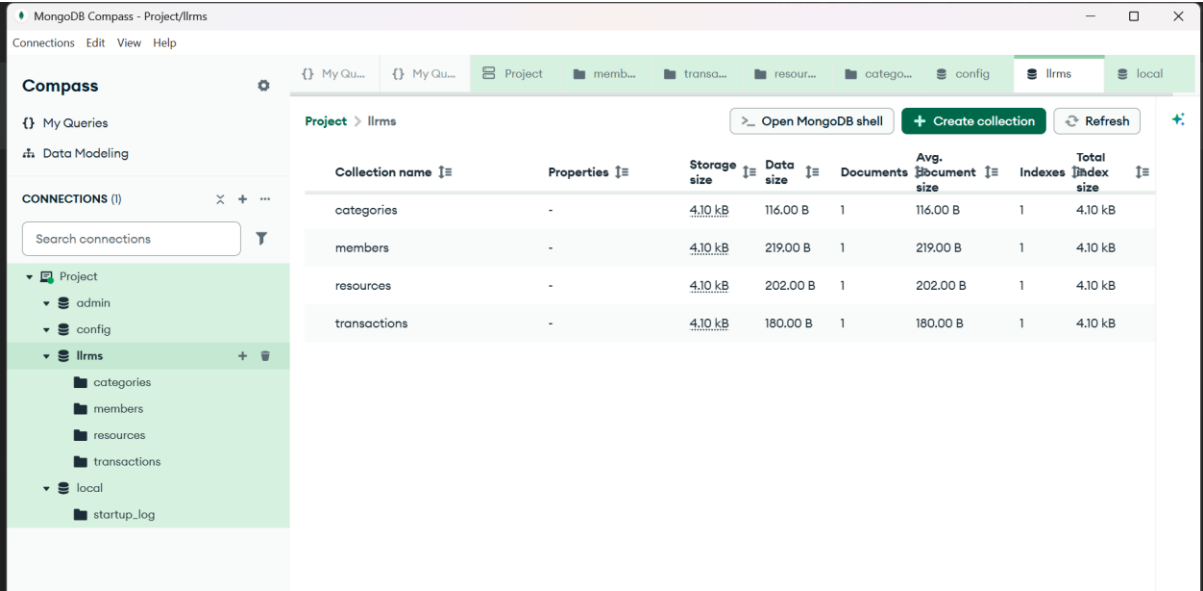


Figure 3.3: Live tracking state of an unreturned book asset within the llrms.transactions bridge collection.

4. llrms collection overview.



5. Phase Conclusion & Next Steps

With the core database definitions properly instantiated, data types verified, and relational parameters validated through the MongoDB Compass GUI workspace, the structural baseline is fully locked. The project is positioned to move to **Phase 2 (Next 25% Scope)**, which will focus on deploying the active Express.js application engine, binding database models via Mongoose schemas, and testing the primary backend REST API server endpoints via Postman.

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